An LED display uses LEDs for backlight and an array of LEDs act as pixels. LEDs emitting white light are used in monochrome (black and white) TV; Red, Green and Blue (RGB) LEDs are used in colour television. The first LED television screen was developed by James P. Mitchell in 1977. It was a monochromatic display. But, after about three decades, in 2009, SONY introduced the first commercial LED Television.

4.12.1 Advantages of LED television

- It has brighter picture quality.
- It is thinner in size.
- It uses less power and consumes very less energy.
- Its life span is more.
- It is more reliable.

Points to Remember

- The magnitude of current is defined as the rate of flow of charges in a conductor.
- The SI unit of electric current is ampere (A).
- The SI unit of electric potential and potential difference is volt (V).
- An electric circuit is a network of electrical components, which forms a continuous and closed path for an electric current to pass through it.
- The parameters of conductors like its length, area of cross-section and material, affect the resistance of the conductor.
- SI unit of electrical resistivity is ohm metre. The resistivity is a constant for a given material.
- The reciprocal of electrical resistivity of a material is called its electrical conductivity.

$$\sigma = \frac{1}{\rho}$$

- The passage of electric current through a wire results in the production of heat. This phenomenon is called heating effect of current.
- One horse power is equal to 746 watts.
- The function of a fuse wire or a MCB is to protect the house hold electrical appliances from excess current due to overloading or a short circuit.

Solved Problems

1. Two bulbs are having the ratings as 60 W, 220 V and 40 W, 220 V respectively. Which one has a greater resistance?

Solution:

Electric power $P = \frac{V^2}{R}$

For the same value of V, R is inversely proportional to P.

Therefore, lesser the power, greater the resistance

Hence, the bulb with 40 W, 220 V rating has a greater resistance.

2. Calculate the current and the resistance of a 100 W, 200 V electric bulb in an electric circuit.

Solution:

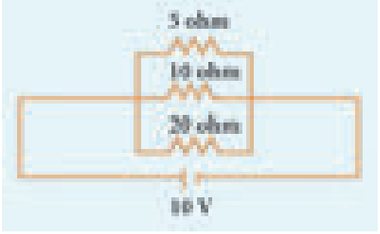
Power P = 100 W and Voltage V = 200 V

Power P = V I

So, Current, $I = \frac{P}{V} = \frac{100}{200} = 0.5$ A

Resistance, $R = \frac{V}{I} = \frac{200}{0.5} = 400\ \Omega$

3. In the circuit diagram given below, three resistors R_1, R_2 and R_3 of 5 Ω, 10 Ω and 20 Ω respectively are connected as shown. Calculate:



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